

Challenges When Co-Designing With Children in the Classroom

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Abstract

Practical insights into running co-design projects with children in school, including common problems that arise and how to manage them, are under-reported. We ran four co-design sessions, each with up to fifteen 10-11 year olds, at a UK primary school. In this paper, we reflect on the co-design process in the classroom and highlight six challenges we faced: school access; unfamiliarity with the school; gaining consent/assent; child engagement; behaviour management; and reporting back our findings to stakeholders. For each challenge, we detail how we mitigated it during the workshops, such as associating assent with specific activities. We also report on further reflections on these issues that occurred after the workshops, facilitated by a focus group with outreach experts and discussions at a workshop with other researchers who co-design with children. The contribution of this paper is to provide practical solutions to challenges faced when co-designing in the classroom.

CCS Concepts

• **Human-centered computing** → **Participatory design**.

Keywords

Co-Design, Classroom, School, Reflection

ACM Reference Format:

Sydney Charitos, Lauren Thompson, Amberly Brigden, Erin Brady, Erin Dooley, Katerina Wintle, and Jon Bird. 2025. Challenges When Co-Designing With Children in the Classroom. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3706673>

Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3706673>

1 Introduction

Involving children in the design of technology that is intended for their use is important as it can lead to the development of technologies that meet their unique needs [8, 20, 24]. Co-design is often used to involve children in the design process, a method closely aligned with participatory design [23] and a core element of user-centred design [1]. Co-design seeks to engage children as design partners, rather than assigning them roles as users, testers, or informants, as outlined in Druin's (2002) framework for categorising children's involvement in the design process [10].

A fundamental challenge with any co-design process is recruiting a representative group of participants. The most prevalent approach used in co-design with children is to recruit them through schools [17], which can provide access to a diverse group of children, particularly those from 'hard to reach' lower socioeconomic groups, who might not otherwise engage in research [4]. Another way to involve children from all socioeconomic groups is by conducting research in their classrooms during the school day, as this means that participation is less dependent on family circumstances, as attendance at activities outside of school hours is often influenced by a family's socioeconomic status. For example, children from less affluent families participate in fewer extracurricular activities [14, 16]. Furthermore, conducting co-design activities in primary schools is also more likely to engage children from diverse socioeconomic backgrounds, as they have lower dropout rates and fewer unauthorised absences than secondary schools in the UK [27] and lower attendance is frequently linked to socioeconomic status [25].

However, practical insights into how to run co-design projects with children in school, such as common problems that arise and how to manage them, remain under-reported [7, 15, 17]. Reporting these 'messy' aspects of research is crucial as it helps to identify the limitations of methodologies and provides pragmatic insights

into how to manage challenges that arise in research [5, 26]. For instance, previous research that has conducted co-design with children in schools has highlighted challenges such as the need to enhance teacher involvement in co-design activities [18], maintain consistent student engagement [9], and adapt content to address diverse learning needs and abilities in classroom contexts [13].

To address this research gap, we reflect on the challenges we faced when we ran four co-design workshops with a group of fifteen 10-11 year olds at a UK primary school. The aim was for the children to create smartwatch visualisations of personal health data to help child personas manage long-term health conditions, such as visualising activity levels for a child experiencing chronic fatigue. We describe six challenges we faced, how we mitigated them in the workshops and our further reflections on them after the workshops that have informed how we will address them in the future.

We make two contributions. First, we provide practical guidance for researchers conducting co-design with children in classroom settings by presenting how we mitigated six challenges that we encountered. This addresses an existing gap in the literature, where there is limited reporting about practical challenges, and how to manage them, when co-designing in the classroom. Second, we highlight the importance of ongoing reflective practice and the value of involving external experts to inform this process and enhance professional development.

2 Methods

2.1 Location and Timings of Workshops

In June 2023, we ran four two-hour co-design workshops in a classroom on consecutive Wednesday afternoons in a primary school in Bristol, UK. The workshops ran during school hours concurrently with school play rehearsals, allowing the children to choose between the two activities. The school was chosen as it had a strong existing partnership with the University of Bristol's Faculty of Engineering Access and Inclusion Team via prior outreach initiatives. A member of the Access and Inclusion Team made the initial introduction to the head teacher and we sent them details of our proposed project. Access to the school for the research project was approved after an in-person discussion. The school is located in a more deprived area of the city (third decile of the English Index of Multiple Deprivation) and has two classes each year for children aged seven to eleven, with 39% being eligible for free school meals. Overall the school has a 'good' rating according to the Office for Standards in Education, Children's Services and Skills in the UK.

2.2 Participants and Recruitment

Before the start of the first workshop, families of final-year students (aged 10-11) received a participant information sheet, a consent form covering all the workshops, and a form to record demographic information. These materials were paper-based rather than digital, as the school advised this was a more accessible format and would help engage more parents. Families wishing to participate completed forms and returned them to the class teacher, who then passed them on to the research team. If demographic forms were incomplete, we sent out one reminder for them to be completed.

In addition to parent/carer consent, we also collected children's assent for each distinct activity that involved data collection. Children were asked to provide written and verbal assent just prior to engaging in these activities in an effort to make their involvement in the process more informed and meaningful [19]. Sometimes children took part in the co-design activities without having parental consent. In these cases, research data was not collected, stored or analysed. Instead, their participation was considered Patient Public Involvement (PPI), which is defined as research conducted with or by members of the public rather than about or for them. The UK's National Institute of Health Research emphasises that PPI data cannot be reported at an individual level but can be presented in terms of general trends or as aggregated data [12]. Up to eight children in each workshop were included in a PPI role. Out of the children who had parental consent (10) and provided demographic information (9), all identified as White British. Four of these children had Special Educational Needs (SEN), two were girls, and the average age was 10.7 years.

2.3 Scheme of Work

The scheme of work across the four workshops was created with support from a teaching practitioner with experience teaching 11-year-olds, who is a co-author of the paper (KW). There were two types of activity in the workshops: creative activities, where children generated their own design ideas; and evaluative activities, where they assessed their own or others' work.

After an initial exploration of their understanding of digital health, children were tasked with creating smartwatch visualisations of personal health data in the first workshop. Later workshops incorporated BBC micro:bits - a child-friendly electronics device with an associated block programming platform [2] - evaluating digital versions of smartwatch visualisations created in earlier workshops, and the creation of a school display in the final workshop. The series of workshops followed a progression from ideation to tangible manifestation, encouraging iterative design thinking and collaboration. The materials generated for workshop one were tested in a pilot with an 11-year-old to ensure the materials were appropriate for our participants. This pilot led to changes in the timing of some workshop activities and the removal of less engaging materials.

2.4 The Reflection Process

We engaged in a four-stage reflection on action process [22] which is outlined below.

Stage 1: Wider Research Team. After each workshop, the researchers who had run the session in the school met with the wider research team to discuss what had happened and identify opportunities for improving future workshops based on these insights. These reflections focused on how the children had engaged with activities and any feedback they had provided.

Stage 2: School Feedback. After the final workshop, we asked the children, class teacher and head teacher for feedback about the project, asking what they thought we did well and what we could have done better. Following their feedback, the research team discussed and categorised the challenges we faced had faced, how

well we had managed to address them during the workshops and what we would do differently in the future.

Stage 3: Outreach Experts. We conducted a two-hour focus group with three outreach experts about the challenges we faced during the workshops. The goal was to learn from their expertise in engaging schools and managing classroom environments as an external organisation to identify strategies to mitigate these challenges. We recruited the participants through the research teams' established professional networks and we refer to them as experts because they currently work in outreach roles within or outside of academia and have experience of collaborating with schools. They also self-reported a high level of outreach knowledge, averaging 9 out of 10. One of the experts (P3) was the member of the Access and Inclusion Team who enabled the initial contact with the school but the other two had not been involved in the project. The focus group, led by two research team members, began with an open discussion about researchers conducting studies in schools. Following this, vignettes were introduced to illustrate examples of the categorised challenges from stage two. Challenges related to consent/assent and reporting findings were not discussed, as the outreach experts were not expected to have expertise in these areas. Insights from the focus group were used to further elaborate on the challenges presented, providing additional context and recommendations for working in schools.

Stage 4: Interaction Design and Children (IDC) Participatory Design Workshop. Using early insights from our work, we submitted a short workshop paper for the ACM IDC 2024 conference workshop 'Participatory Design with Young Children: Failures, Challenges, and Successes'. We were able to discuss our experience of doing co-design with children in the classroom with other attendees. The workshop led to a recognition that reporting the challenges we had faced, how we mitigated them and further reflections for future practice would be useful for this research community.

3 Challenges, Mitigations and Further Reflections

Our reflection led to the identification of six challenges: school access; unfamiliarity with the school; obtaining consent/assent; child engagement; behaviour management; and reporting findings to stakeholders. These challenge areas are reported in the order of when they occurred during the co-design project. We detail how we mitigated each challenge during the workshops. Engagement and behaviour management challenges are grouped under 'Workshop Facilitation' due to shared mitigation strategies. We provide further reflections on how we would address the challenges differently in the future, which incorporate insights from the outreach experts and discussions with workshop participants.

3.1 School Access

Challenge. The main challenge in accessing a school to conduct this research was the overwhelming burden these institutions already face. Educational institutions often receive proposals from external organisations to do activities with their students. Given the substantial demands already placed on teachers' time and resources,

their ability to thoroughly evaluate and choose appropriate activities from this extensive pool is often limited. Furthermore, teachers may not have the capacity to support activities due to the additional administrative workload this would entail. In this context, existing relationships with schools are invaluable, and individuals or groups that have cultivated these connections carefully protect them due to the challenges of establishing and maintaining such relationships and concerns about potential misuse.

Mitigation. The key mitigation strategy was to present our co-design project in a way that met the different stakeholders' needs and expectations. When approaching the Access and Inclusion Team about gaining access to a school, it was important to foster their trust in our ability as researchers to manage such a significant relationship. Building this trust involved: engaging members of the team in detailed discussions about the project and incorporating their feedback; participating in other events they led to demonstrate that we could run activities with children in a classroom; and committing to sharing findings with their team.

When we established a relationship with the Access and Inclusion Team, they introduced us to the school, with whom they had an established connection, and this was invaluable for getting a meeting with the head teacher. We also gained valuable insights from the Access and Inclusion Team about how to effectively communicate with schools by understanding their needs and expectations. They emphasised the importance of moving at the school's pace, making the benefits of a project clear, and minimising any administrative burden for the school.

When approaching the head teacher, we presented the research as a low-risk, self-contained workshop series with clear benefits which aligned with previous outreach work that the school had previously approved. We provided all participant-facing documentation upfront, ensuring that it was clear that all materials were prepared and that minimal administrative input would be required. We suggested our research could take place after the end of an exam period, just before the summer holidays, during a time when the school timetable was more flexible. We also made clear that our only request was for a teacher to be present in the room to provide behavioural support.

Further Reflection. When attempting to build a relationship with a school, it is crucial to recognise that primary schools often view this as establishing a connection with the university rather than with a specific research group. As one expert noted, "we all know that a university is not one thing, but when you're sitting in a primary school, the university is one thing" (P2). Therefore, future work should prioritise shifting the focus from viewing research with a school as a one-off transaction to recognising it as an ongoing relationship. As one expert explained, instead of portraying a project as "a week's work" (P1), researchers should recognise that these initiatives often require a longer time frame e.g. working with a school intermittently for 9 to 12 months. Each project must also try to minimise the administrative burden for schools while maximising benefits for the children. One such benefit, which we did not emphasise enough in this project, was aligning our activities with the Primary National Curriculum [11].

An effective approach to establishing a new relationship with a school is to leverage an existing one. In this research project,

we utilised an established relationship between the school and the university, and, more specifically, P3. However, this strategy is not without its challenges. As our key point of contact articulated, “It feels extremely risky for me as the person that’s garnered this long-term relationship. [...] It’s really risky, risking something I’ve spent years and years building” (P3). This highlights the need to invest in relationships with those who have access to schools rather than viewing them simply as a resource. Researchers must also recognise that gatekeepers may decline requests, as it is their responsibility to protect the relationships they have established as “you can’t then send 2000 researchers throughout each year, otherwise you create the same problem of them being completely overwhelmed with offers” (P1).

3.2 Unfamiliarity with the School

Challenge. A key challenge was our unfamiliarity with the school, which resulted in uncertainty about a number of important details before the workshops began. For example, we did not know how many children would participate in the workshops, the specific classroom we would use, whether the class teacher was available for behavioural support and whether any of the children had additional learning needs, neurodiversity, or long-term health conditions that might influence their engagement with the workshops. This unfamiliarity was compounded by us wanting to minimise any administrative burden on the school, leading us to hesitate in repeatedly requesting information as decisions were being made.

Mitigation. The research team managed this uncertainty in two ways: preparing flexible age-appropriate materials; and making the most of any opportunities to gain familiarity with the school environment. Firstly, we developed a flexible scheme of work that allowed us to add or modify elements based on the initially unknown time frame we had to deliver our workshops. Additionally, the materials for the first workshop were tested in a pilot study with a Year Six child and a teaching practitioner (KW) to ensure they were appropriate for the age group.

Secondly, to increase our familiarity with the school environment, two researchers who facilitated the workshops attended two outreach sessions involving micro:bits at the same school (totalling four hours), one month prior to the co-design workshops. The benefits of this familiarisation included identifying suitable areas for setting up a display that was produced in workshop four to showcase the children’s work; learning the school’s rules, such as the prohibition on phones, which helped the researchers demonstrate respect for and adherence to these guidelines; and understanding the classroom environment, including the layout of desks and the equipment available for use, such as a whiteboard or smart board.

However, it is important to note that this familiarisation process may have led to researchers having incorrect expectations. During workshop three, the researchers assumed the children had prior experience with micro:bits, based on observations from earlier outreach sessions and being told by a teacher that they used micro:bits in their IT classes. This assumption proved incorrect, as many children did not have this basic experience with the micro:bit platform and seemed very unfamiliar with the device.

Further Reflection. We did not effectively mitigate our uncertainty and faced numerous unknowns prior to the first workshop, particularly concerning the children who would be involved and the class teacher. This may be a consequence of organising the project within over-burdened schools, where organising projects with external organisations is not one of their priority tasks. Therefore, rather than expecting the school to provide this information, a more effective approach is to prepare materials to ensure accessibility. For instance, in future we could align our materials better with Quality First Teaching practices, which emphasise high expectations alongside universally accessible activities [6], thereby maintaining an inclusive approach throughout the project.

This approach also underscores the importance of asking, “where are [the children] starting from?” (P2), ensuring that research activities are designed with children’s perspectives in mind rather than expecting them to adapt to the researchers’ requirements. Outreach experts highlighted examples where children’s needs may not be adequately addressed in research, such as using overly complex activities, conducting workshops that cater to a wide range of age groups without appropriate differentiation, or failing to consider how to motivate children to engage with the research activity.

3.3 Consent and Assent Process

Challenge. There were two key challenges in the consent and assent process. Firstly, obtaining assent was more difficult than anticipated due to the children’s group dynamics and their concerns regarding photography. For instance, in the final workshop, we organised children into rotating groups and collectively asked them to sign assent forms. Despite some children having provided individual assent in previous weeks, vocal children often influenced their groups to decide against assenting to share their contributions. Additionally, many children did not like the idea of having their picture taken. This led to an all-or-nothing mentality, with some children declining to assent to any aspect of the activity because they did not want to be photographed.

The second challenge was managing the involvement of two groups of children: those who had consent to be study participants and those who participated in a PPI role. Since the research took place during school hours, children could choose to engage in the study in a PPI capacity without parental consent, but as noted in Section 2.2, this meant we could not collect and report on research data from these children. This led to the challenge of ensuring that those without consent were not captured on recording devices and that their contributions were not mistakenly grouped with the data from those with consent. The situation was further complicated by fluctuating attendance due to school absences, movement between the workshops and school play rehearsals, and some children receiving parental consent for participation in later workshops.

Mitigation. To mitigate the challenge of obtaining assent, in later workshops we took assent individually and linked it to a specific contribution, effectively reducing the impact of group dynamics and all-or-nothing thinking regarding photography. For instance, researchers held up something a child had created and explicitly asked them whether they were comfortable sharing it, rather than asking them to give assent to share something they hadn’t yet created. This meant that the children knew what they were assenting

to and could make an informed judgement about whether they were happy or not to share it. There were still many occasions where the children chose not to assent to sharing their creations, demonstrating their autonomy to make decisions about sharing their contributions.

We managed the challenge of working with a group of research participants and a group in a PPI role by separating the two groups into two locations in the classroom. This ensured that their contributions could be collected separately and that the audio recordings would not capture the voices of non-consenting children. However, this separation caused some conflict, as friends often wanted to sit together. For instance, one child explicitly told the researchers that they asked their parent/carer to provide consent solely so that they could sit with their friend in the consent group. To address this problem, we introduced activities in the final workshop that did not separate the children, reducing the impact of group division and allowing for more inclusive participation. However, this adjustment meant that the final activity could not be reported in a research context.

Further Reflection. At IDC 2024, Read et al. introduced their ‘save or share’ principle [21], which helps children decide what they might want to ‘save’ for themselves and what they want to ‘share’ with researchers. This approach emphasises the importance of continuously asking for assent rather than asking a child to provide assent once for an entire project. To enhance our approach in the future, we could incorporate Read et al.’s tangible and engaging approach to collecting assent, which in their case was a large red post-box where children posted work they were happy to share.

3.4 Workshop Facilitation

Challenge: Engagement. We had many challenges with keeping the children engaged in the workshop activities, leading to them often telling us that they were bored. This challenge can be broken down into three areas: the content of the activity; activity timing; and continuity. Regarding engagement with activity content, there were certain activities the children disengaged from. For instance, children were often reluctant to do drawing activities and some children chose to outsource the drawing to friends or researchers, citing a lack of confidence in their own skills. There was also disengagement from the micro:bit activity due to researchers’ incorrect assumptions about the children’s skill levels (see Section 3.2), meaning that the activities were too difficult for some children.

Activity timing was another challenge exacerbated by the researchers’ lack of experience in classroom settings and assumptions from the pilot evaluation of the workshop materials. The pilot, conducted with a single child and researcher, did not accurately reflect how long it would take the larger group of children to complete activities. As researchers adapted in the moment and became more flexible with timings, confusion and disengagement arose, with children expressing a desire for clearer completion timelines and more explicit breaks.

Maintaining the continuity of the project was also a challenge, with the children forgetting what they had created between workshops. Despite providing recaps at the beginning of each workshop, some children still struggled to grasp both what and why the research was happening. This issue was exacerbated by fluctuating

attendance, resulting in varying levels of familiarity and experience among the children in the classroom. Consequently, fostering a cohesive group dynamic became increasingly difficult, as varied levels of familiarity and experience resulted in differing levels of participation and investment in the project.

Challenge: Behaviour Management. As researchers facilitating the workshops had limited experience in primary school classroom teaching, we struggled to navigate the established interpersonal relationships among Year Six students, which often led to challenging group dynamics during creative tasks. The children’s behaviour during workshops aligns with the framework outlined by Van Mechelen et al. [29], which identifies six challenging group dynamics in participatory design with children: ‘groupthink’, where the group is less capable than its individual members; ‘destructive conflict’, where disagreements lead to division within the team; ‘free riding’, where some children contribute minimally, relying on others’ efforts; ‘apart together’, where group contributions consist of disconnected individual ideas; ‘unequal power’, where certain group members exert disproportionate control over decisions; and ‘laughing out loud’, where a playful atmosphere reduces task engagement.

Groupthink posed significant challenges, as illustrated in workshop three where a boy initially liked a unicorn design but later altered his preference to align with the group’s dislike for it. Destructive conflict occurred during a cutting and glueing activity in workshop three, where pairs competed for roles of the ‘the cutter’ or ‘the gluer’, leading to a refusal to collaborate and significant delays. Concurrently, free riding was evident as some participants relied on their partners to be ‘the cutter’ and ‘the gluer’. A significant challenge arose during the final workshop, as groups were expected to collaboratively develop a design idea. However, the ‘apart together’ and ‘unequal power’ dynamics disrupted this process and the children struggled to create a cohesive design. Laughing out loud was another disruptive dynamic throughout the workshops, with some children favouring making jokes with their friends over completing the task. This behaviour often influenced nearby groups, resulting in major disruptions. We also experienced a challenging group behaviour not mentioned by Van Mechelen et al., when in some of the workshops a ‘bystander’ group emerged, characterised by passivity and a lack of motivation to get involved in activities. This subset of children often remained idle until a researcher intervened.

Mitigation. We mitigated these challenges by reflecting on mistakes, building on successes and utilising the experience of the class teacher present. The research team engaged in reflective practice after each workshop to discuss improvements to how we facilitated them. For instance, after reflecting on issues related to flexible activity duration, we introduced more defined timings with structured breaks and communicated these expectations clearly to the children. Both the children and the class teacher, in their feedback form, responded positively to the increased structure. We explored various strategies to address the lack of continuity in the project, focusing on helping the children who were struggling to understand the research aims. We rephrased the research goals, used a variety of visuals to communicate them to children and engaged in more conversations to reinforce why their input was valuable.

In terms of building on successes, as the research team became more familiar with the group, they identified individuals requiring closer attention to help them stay on task, which enhanced the running of the workshops and group work. Additionally, having team members with teaching or facilitation experience was useful for adapting content and managing classroom dynamics in the moment. For example, when the ‘bystander’ group lost focus, a researcher intervened to redefine the activity’s goals and re-establish the project’s context. In another instance, when a child began drawing on another’s worksheet, instead of reprimanding them, the researcher employed positive redirection, borrowing the pen and providing an alternative task. Such strategies, rooted in Quality First Teaching principles, helped de-escalate situations and maintained a constructive environment [6].

There were two main areas where the experience of the class teacher was used to support the facilitation of the workshops. Firstly, when behaviour became disruptive, the teacher was able to intervene, leveraging their existing relationships to de-escalate situations and maintain focus. Secondly, in later workshops we asked the class teacher for support in grouping children that would work well together, informed by their in-depth knowledge of class dynamics.

Further Reflection. Although the teacher occasionally intervened to help manage the children’s behaviour, for most of the workshops, they remained in a more peripheral role, using this time to focus on other tasks, such as marking. Initially, we interpreted this as a positive outcome, aligning with our goal to minimise the burden on school staff (as discussed in Section 3.1). However, the outreach experts highlighted that we were too concerned that the class teacher might view their involvement as a burden. One outreach expert noted, “you know, they really love it; they enjoy their jobs. When it involves doing stuff with the kids, that’s what motivates them” (P2). Therefore in future, we would aim to position the class teacher as a co-designer in the research, as their involvement can enrich the workshops, enabling researchers to benefit from their pedagogical insights and their established rapport with the children. Further, teachers can also provide a sense of continuity and ensure the project is connected to the broader classroom context rather than being perceived as solely linked to the external research team. For example, in workshop one, the class teacher (un-prompted) motivated the children by acknowledging their work and awarding house points for their efforts.

The outreach experts’ reflections on challenging group dynamics provided valuable insights into our ‘laughing out loud’ and ‘bystander’ groups. They suggested the ‘laughing out loud’ group likely stemmed from the research team’s attempts to foster an informal environment by wearing casual clothing and encouraging creativity. Consequently, this informality may have resulted in more disruptions and fewer shared ideas, rather than fostering a safe space for expression. The outreach experts noted that some disruption may be unavoidable in efforts to foster creativity and autonomy. However, they cautioned that disrupting the established school structure could undermine the consistency of the children’s day, potentially negatively impacting their workshop experience and causing further disruptions, especially for children with SEN.

Additionally, the emergence of the ‘bystander’ group may reflect the children’s lack of opportunities to experience failure, which can

limit their resilience and lead to hesitance when faced with open-ended questions. This underscores the importance of researchers creating opportunities for children to engage with uncertainty, as although the unknown can be challenging to navigate, it also offers space for exploration and learning.

Therefore, in future workshops, we will establish behavioural ground rules from the outset, ideally aligning them with the existing classroom rules that the children know and respect. We could then gradually transition from topics the children are confident discussing to more open-ended and creative discussions as the workshops progress.

3.5 Reporting Back

Challenge. One of the major challenges faced during the project was reporting back findings to the children who were involved in the study. This difficulty was compounded by the fact that the workshops were held at the end of the academic year, which meant there was not much time to analyse their data and then report findings back to them before the summer holidays.

Mitigation. To address this challenge, the research team incorporated multiple feedback mechanisms into the project design. This included providing immediate verbal feedback at the end of each workshop, where researchers highlighted the children’s contributions and discussed how their ideas would influence the next stage of the project. Additionally, the final workshop focused on consolidating all the work completed over the previous weeks and creating a display for the school’s reception area. The nature of this display shifted from a poster to a tangible installation, making the final summary more engaging for the children. Further, by having the display in a public area of the school, their work could be showcased to other students, teachers, and potentially their parents.

The research team also shared the workshop materials and the feedback from the children and school with the Access and Inclusion Team, ensuring that subsequent initiatives could benefit from the lessons learned during this research.

Further Reflection. As noted by the outreach experts, “they are not just individual child units; they come with brothers, sisters, mums, dads, uncles, aunties, and random adults... [...] you’re building up this relationship with someone’s auntie as much as you are with that individual child” (P2). Consequently, when planning the output of research, it is important to consider what the children are taking home or sharing with their wider networks. By creating shareable outputs, researchers can build positive relationships with these networks, which can benefit children by encouraging their networks to recognise their contributions.

4 Final Reflections

Reflecting on the workshops led to us identifying six main challenges, summarised in Figure 1, which vary in when they need to be addressed and how difficult they are to mitigate. Some challenges, such as school access, unfamiliarity with the school, and reporting findings to stakeholders, were primarily encountered during the planning and logistical stages of the project. These challenges required relationship-building with the Access and Inclusion Team

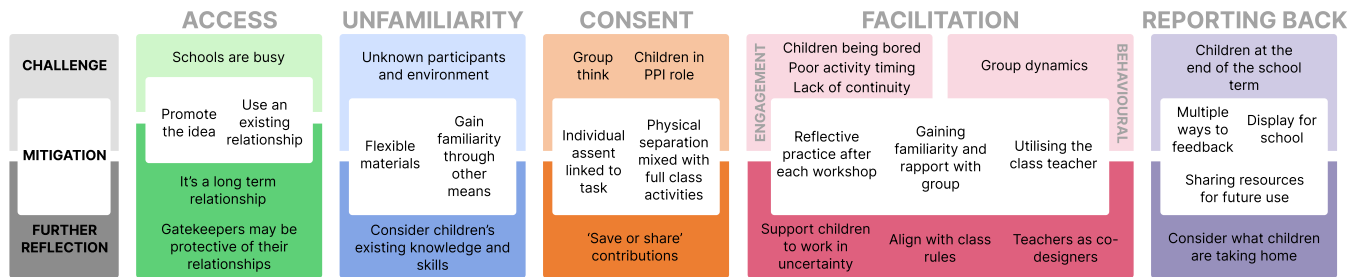


Figure 1: Overview of the challenges, mitigations, and further reflections identified in the project.

and the school involved in the study. In contrast, challenges related to obtaining consent/assent, child engagement, and behaviour management arose predominantly during the workshops. These issues demanded real-time adaptations and were highly sensitive to contextual factors, such as fluctuating attendance, varying levels of interest, and even external influences like hot weather, which made it harder for the children to concentrate.

These challenges were mitigated in a variety of practical ways, such as preparing flexible age-appropriate materials, linking assent to a specific contribution, and making the research output concrete and shareable. There were also mitigation strategies that required a greater level of collaboration with the school, such as presenting research in a way that meets different stakeholder needs to gain access, using existing connections to gain familiarity with the school environment and utilising the experience of the class teacher. These strategies emphasise the importance of relationships when conducting research within a school setting, as strong connections not only facilitate access but also enrich the research experience for all stakeholders, ensuring that it is more responsive to the needs of both students and educators.

By documenting our experience and reflections in this case study, we aim to address an existing research gap by providing practical insights into how to run co-design workshops with children in schools. We offer actionable strategies to assist researchers in overcoming the common challenges of conducting co-design in the primary school classroom. These insights, particularly strategies for mitigating challenges during the planning and logistical stages, could also be generalised to secondary school settings, making the guidance broadly applicable across various educational levels. Furthermore, we have demonstrated the value of extensive reflection, both within the research team and with external experts, for gaining a better understanding of how we can effectively co-design with children in the classroom [3, 22, 28].

Acknowledgments

This work was supported by the Engineering and Physical Sciences Research Council [Grant Number: EP/S100597], the Faculty of Engineering, Access and Inclusion Team, University of Bristol and the Jean Golding Institute, University of Bristol. Many thanks to the children and teaching staff who participated in and supported this project as well as the outreach experts for their invaluable feedback.

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